

1. Gaussian Amplitudes

a) Compute the normalisation constant A

For a Gaussian amplitude, the normalization is

$$\int_{-\infty}^{\infty} |G(x)|^2 dx = 1$$

Substituting the given form for $G(x, t)$ we get,

$$\int_{-\infty}^{\infty} \left(A e^{-\frac{(x-ct)^2}{4a^2}} e^{i(k_0x - \omega_0t)} \right)^2 dx = 1$$

We can separate the expression from the phase term:

$$\begin{aligned} \left(A e^{-\frac{(x-ct)^2}{4a^2}} e^{i(k_0x - \omega_0t)} \right)^2 &= A^2 e^{-\frac{(x-ct)^2}{2a^2}} e^{i(2k_0x - 2\omega_0t)} \\ e^{i(2k_0x - 2\omega_0t)} A^2 \int_{-\infty}^{\infty} e^{-\frac{(x-ct)^2}{2a^2}} dx &= 1 \end{aligned}$$

As the phase term doesn't contribute to the normalization, we can omit them from the calculations,

$$A^2 \int_{-\infty}^{\infty} e^{-\frac{(x-ct)^2}{2a^2}} dx = 1$$

Using the property of the Gaussian integral,

$$\int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \sqrt{\frac{\pi}{a}}$$

We get,

$$A^2 \sqrt{2\pi a^2} = 1$$

Solve for A

$$A^2 = \frac{1}{\sqrt{2\pi a^2}}$$

Or,

$$A = \frac{1}{\sqrt{\sqrt{2\pi a^2}}}$$

b) Show that the centre of the wave packet, i.e., the average position $x_0(t) = \langle x_0 \rangle(t)$, moves at the speed of light c .

Using the given definition of the spatial average of a field, we get

$$\begin{aligned}\langle x \rangle(t) &= \int_{-\infty}^{\infty} x \left(A e^{-\frac{(x-ct)^2}{4a^2}} e^{i(k_0 x - \omega_0 t)} \right)^2 dx \\ &= A^2 \int_{-\infty}^{\infty} x e^{-\frac{(x-ct)^2}{2a^2}} dx\end{aligned}$$

We can compute the integral using integration by parts,

$$\begin{aligned}&= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}a} e^{-\frac{(x-ct)^2}{2a^2}} dx \\ &= \int_{-\infty}^{\infty} [(x-ct) + ct] \frac{1}{\sqrt{2\pi}a} e^{-\frac{(x-ct)^2}{2a^2}} dx \\ \langle x \rangle &= \frac{1}{\sqrt{2\pi}a} \int_{-\infty}^{\infty} (x-ct) e^{-\frac{(x-ct)^2}{2a^2}} dx + \frac{ct}{\sqrt{2\pi}a} \int_{-\infty}^{\infty} e^{-\frac{(x-ct)^2}{2a^2}} dx\end{aligned}$$

let $w = x - ct$

$$= \frac{1}{\sqrt{2\pi}a} \int_{-\infty}^{\infty} w e^{-\frac{w^2}{2a^2}} dw + ct \int_{-\infty}^{\infty} |G(x)|^2 dx$$

The first term amounts to zero because the integral is even but the term itself is odd, and for the second term we get that its equal to 1 from normalization

$$\langle x \rangle = ct = x_0$$

Q.E.D.

c) Prove that the Gaussian wave packet does not disperse by showing that the variation $\langle (x - x_0)^2 \rangle(t)$ is independent of time

Again using the definition given in the problem statement, we have

$$\begin{aligned}\langle (x - x_0)^2 \rangle(t) &= \int_{-\infty}^{\infty} (x - x_0) \left(A e^{-\frac{(x-ct)^2}{4a^2}} e^{i(k_0 x - \omega_0 t)} \right)^2 dx \\ &= \frac{1}{\sqrt{2\pi}a} \int_{-\infty}^{\infty} (x - ct)^2 e^{-\frac{(x-ct)^2}{2a^2}} dx\end{aligned}$$

Let $w = x - ct$

$$\begin{aligned}&= \frac{1}{\sqrt{2\pi}a} \int_{-\infty}^{\infty} w^2 e^{-\frac{w^2}{2a^2}} dw \\ \frac{d}{dw} e^{-\frac{w^2}{2a^2}} &= e^{-\frac{w^2}{2a^2}} \frac{d}{dw} \left(-\frac{w^2}{2a^2} \right) \\ &= -\frac{2w}{2a^2} e^{-\frac{w^2}{2a^2}}\end{aligned}$$

$$= -\frac{w}{a^2} e^{-\frac{w^2}{2a^2}}$$

The integral becomes,

$$= -\frac{a}{\sqrt{2\pi}} \int_{-\infty}^{\infty} w(e^{-\frac{w^2}{2a^2}}) dw$$

$$\langle (x - ct)^2 \rangle = -\frac{a}{\sqrt{2\pi}} \int_{-\infty}^{\infty} w(e^{-\frac{w^2}{2a^2}}) dw$$

Integration by parts

$$\frac{d}{dx}(uv) = u'v + uv'$$

$$\Rightarrow uv' = \frac{d}{dx}(uv) - u'v$$

$$\int \Rightarrow \int dx uv' = \int dx \frac{d}{dx} uv - \int dx u'v$$

$$\int dx uv' = uv - \int dx u'v$$

Hence,

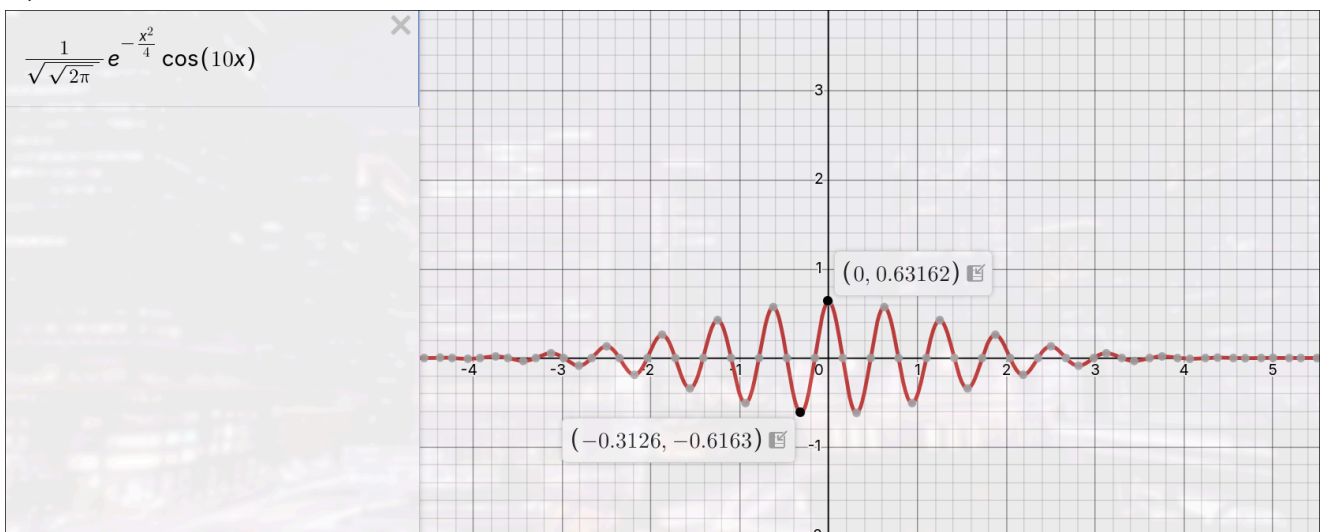
$$\langle (x - x_0)^2 \rangle = -\frac{a}{\sqrt{2\pi}} w e^{-\frac{w^2}{2a^2}} \Big|_{-\infty}^{\infty} + \frac{a}{\sqrt{2\pi}} \int dx e^{-\frac{w^2}{2a^2}}$$

$$= 0 + a^2 \int du \frac{1}{\sqrt{2\pi a}} e^{-\frac{w^2}{2a^2}} = a^2$$

which is a constant that is independent of time.

2. Plots of Gaussian Waves

a)



b)

